

How to Write Chapter 1: Introduction

A Step-by-Step Guide for Undergraduate Statistics Projects

Issah Seidu (PhD)



Department of Statistics and Actuarial Science

Outline

- ① Overview of Chapter 1
- ② Cover Page & Title Page
- ③ 1. Background of the Study
- ④ 2. Problem Statement
- ⑤ 3. Objectives
- ⑥ 4. Significance of the Study
- ⑦ 5. Methodology (Overview)

What is Chapter 1?

Purpose

Chapter 1 is the **gateway** to your entire project. It tells the reader:

- **What** you studied
- **Why** it matters
- **How** you studied it
- **Who** benefits

Sections of Chapter 1

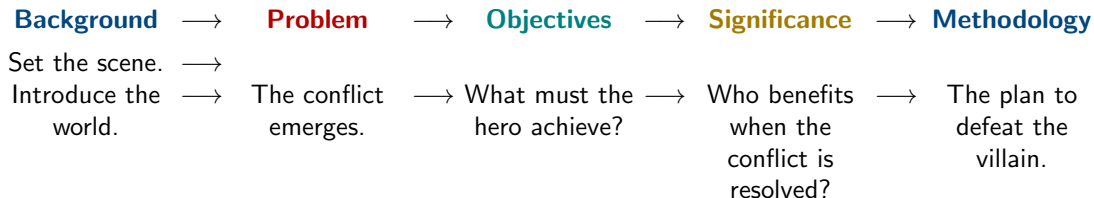
- ① Background of the Study
- ② Problem Statement
- ③ Objectives
- ④ Significance of Study
- ⑤ Methodology (Overview)
- ⑥ Organisation of the Study

Key Rule

A well-written Chapter 1 makes the reader *want* to read the rest of your work.

Think of Your Research Like a Movie

*“Writing research is like writing a movie. You lay the background of both the protagonist and the antagonist — which leads to the **problem**, then the **objectives**, then the **significance**, then how the problem will be solved through the **methodology**.”*



So...

Every section of Chapter 1 has a **role to play**. If a section feels disconnected, you have lost your audience — just like a scene that does not advance the plot.

Your Cover Page — Getting It Right

Do NOT forget your Cover / Title Page!

Every project must begin with a properly formatted cover page.

Must-Have Elements

- Full title of the project
- Department & University name
- Course name & course code
- **All group members' full names**
- **All group members' ID numbers**
- Name of supervisor
- Submission date / semester

Example Title Page Layout

University of XYZ

Department of Statistics

**The Effect of Inflation on
Household Expenditure in Ghana**

*Submitted in partial fulfilment of
STAT 400*

Name	ID
Ama Owusu	20201001
Kofi Mensah	20201045

Section 1 — Background of the Study

What is it?

Provides the **context and setting** of your research. It connects global knowledge to the local or specific problem you are addressing.

What to include — move from **BROAD** to **NARROW**:

- ① **Global context:** What is known worldwide about this topic?
- ② **Regional/National context:** What is the situation in your country/region?
- ③ **Local/Specific context:** What gap or situation motivates your study?
- ④ **Justify with evidence:** Use **cited statistics, reports, and literature**.

Common Mistakes

- Writing opinions without citations
- Starting too narrowly (jumping straight to the problem)
- Copy-pasting text without paraphrasing and citing

Background of the Study — Example Structure

Worked Example (Inflation Study)

Paragraph 1 (Global): “Inflation is a persistent macroeconomic challenge affecting economies worldwide. According to the **International Monetary Fund (IMF, 2023)**, global inflation averaged 8.7% in 2022...”

Paragraph 2 (Sub-Saharan Africa): “In Sub-Saharan Africa, rising food and energy prices have exacerbated household vulnerabilities (**World Bank, 2023**)...”

Paragraph 3 (Ghana — specific): “Ghana recorded an inflation rate of 54.1% in December 2022, the highest in three decades (**Ghana Statistical Service [GSS], 2023**), severely impacting the purchasing power of households...”

Paragraph 4 (Narrowing to the gap): “Despite the severity, few statistical studies have modelled the effect of inflation on household expenditure patterns in the Ashanti Region. This study addresses this gap...”

Notice: Every factual claim has a citation. The funnel goes from global → national → local → gap.

Section 2 — Problem Statement

What is it?

A clear, concise statement of the **specific problem** your study addresses. It should naturally follow from the background.

A good problem statement answers:

- What is the **problem**?
- **Who** is affected?
- What is the **consequence** of the problem if unresolved?
- What **knowledge gap** does this study fill?

Do NOT

- State objectives here
- Be vague: “There are many problems with. . .”
- Write more than one page

Do

- Be specific and focused
- Use cited evidence to justify the problem
- End with what the study will do about it

Problem Statement — Example

Worked Example

“Despite the sharp rise in inflation in Ghana, relatively little empirical work has quantified its statistical association with household expenditure at the disaggregated (regional) level (**Asante & Boateng, 2021; GSS, 2022**). Existing national-level analyses mask regional heterogeneity, making it difficult for policymakers to design targeted interventions (**Owusu et al., 2020**). Without region-specific statistical models, households and planners lack an evidence base to anticipate expenditure changes in response to inflationary shocks. This study, therefore, investigates the effect of inflation on household expenditure in the Ashanti Region of Ghana using multiple regression analysis.”

What this paragraph does well:

- Identifies the gap
- Explains the consequence
- Ends with what the study will do
- Every claim is cited

Tip

The last sentence of your problem statement often **directly echoes** your main objective.

Section 3 — Objectives

Main Objective

- **One sentence only**
- Broad overarching aim of the study
- Begins with an **infinitive verb**:
 “To investigate. . .”
 “To examine. . .”
 “To analyse. . .”
 “To assess. . .”
- Should mirror the title of your project

Specific Objectives (3–5)

- Break the main objective into **measurable, achievable steps**
- Each begins with an action verb
- Each maps to a **chapter, method, or result** in your study
- Typically **3 to 5** specific objectives

Golden Rule

Every specific objective must be **answerable** by the end of your project. If you cannot answer it, remove or reframe it.

Objectives — Example

Main Objective

“The main objective of this study is **to investigate the effect of inflation on household expenditure** in the Ashanti Region of Ghana.”

Specific Objectives

- ➊ To **describe** the socio-demographic characteristics of households in the Ashanti Region.
- ➋ To **examine** the trend in household expenditure over the study period.
- ➌ To **determine** the relationship between inflation and household expenditure.
- ➍ To **develop** a regression model for predicting household expenditure from inflationary indicators.
- ➎ To **identify** demographic factors that moderate the effect of inflation on household expenditure.

Notice: Each objective uses a different action verb and is clearly measurable.

Useful Action Verbs for Objectives

Level	Verbs
Descriptive	To describe, To summarise, To characterise, To profile, To document
Exploratory	To examine, To explore, To identify, To determine
Analytical	To analyse, To compare, To assess, To evaluate, To measure
Modelling	To develop, To build, To estimate, To fit, To forecast
Inferential	To test, To investigate, To establish, To ascertain

Avoid

Vague verbs like *“to look at”*, *“to understand”*, or *“to study”* — they are not measurable!

Section 4 — Significance of the Study

What is it?

Explains **who benefits** from your study and **how**. Justify *why the world is better* because you did this research.

Structure: Address each beneficiary/party directly in a bullet point.

Example — Inflation & Household Expenditure Study

- **Policymakers and Government:** Findings provide empirical evidence to guide the design of targeted fiscal and monetary interventions that cushion households from inflationary shocks.
- **Households and Consumers:** The study enhances awareness of how inflation affects spending power, enabling better personal financial planning.
- **Researchers and Academics:** The regression model and dataset contribute to the growing body of literature on inflation dynamics in developing economies.
- **Non-Governmental Organisations (NGOs):** Results can direct social support

Significance of the Study — Tips

Common Mistakes

- Generic statements: “This study is important because. . .”
- Listing parties without saying **how** they benefit
- Repeating the objectives
- Not connecting the benefit to your specific findings

Best Practices

- Name each party **in bold** as a bullet header
- Be **specific**: say what the benefit is
- Consider: practitioners, policymakers, academics, communities, and future researchers
- 4–6 bullet points is usually sufficient

Section 5 — Methodology (Brief Overview in Chapter 1)

Purpose in Chapter 1

This is a **brief overview** — 3 paragraphs only. The full, detailed methodology belongs in **Chapter 3**. Here, you give the reader enough context to understand how the study was conducted.



Methodology Paragraph 1 — Data

If Using **Secondary** Data

State:

- **Source** (e.g., Ghana Statistical Service)
- **Owner/Organisation** that collected it
- **Medium** (database, report, website, CD)
- **Duration/Period** covered (e.g., 2010–2022)
- **Variables** of interest

If Using **Primary** Data

State:

- **Research design** (cross-sectional, longitudinal, experimental)
- **Population** (who/what you are studying)
- **Sampling method** (simple random, stratified, systematic, convenience, etc.)
- **Sample size**
- **Instrument** (questionnaire, interview guide, observation checklist)

Secondary Data Example

“This study used secondary data obtained from the **Ghana Living Standards Survey Round 7**

Methodology Paragraph 2 — Statistical Methods

What to include

Briefly mention the **three tiers** of statistical analysis used (where applicable):

- ① **Descriptive statistics** — to summarise and describe the data
- ② **Inferential statistics** — to test hypotheses and make generalisations
- ③ **Statistical models** — to explain relationships or make predictions

Example Paragraph 2

“Descriptive statistics, including **means, standard deviations, frequencies, and percentages**, were used to summarise the characteristics of the study sample. **Pearson’s correlation analysis** and **independent samples *t*-tests** were employed to examine relationships and group differences respectively. A **multiple linear regression model** was fitted to determine the effect of inflation on household expenditure while controlling for household size, income, and location. Model diagnostics, including tests for **multicollinearity, heteroscedasticity, and normality of residuals**, were conducted to validate the model.”

Methodology Paragraph 3 — Statistical Software

What to include

- **Name** of the software
- **Version** number
- **Specific packages/libraries** used (if applicable)
- Why that software was chosen (optional, one line)

Example Paragraph 3

“All statistical analyses were performed using **R version 4.3.1** (R Core Team, 2023). The tidyverse package (Wickham et al., 2019) was used for data manipulation and visualisation, car (Fox & Weisberg, 2019) for regression diagnostics, and ggplot2 (Wickham, 2016) for graphical displays. Where applicable, **SPSS version 29** (IBM Corp., 2022) was used as a supplementary tool for cross-tabulation and frequency analysis.”

Important

Software must be **cited** just like any other reference! R. SPSS. Stata. Python — all have

Section 6 — Organisation of the Study

What is it?

A short section (typically one paragraph **or** a structured list) telling the reader what each chapter of the project contains.

Example — 5-Chapter Structure

Chapter One introduces the study by providing the background, problem statement, objectives, significance, a brief overview of methodology, and the organisation of the study.

Chapter Two presents a review of relevant literature, covering theoretical frameworks and empirical studies related to inflation and household expenditure.

Chapter Three details the methodology, including the research design, data sources, sampling procedures, and analytical techniques employed.

Chapter Four presents and discusses the results of the data analysis in accordance with the stated objectives.

Chapter Five provides a summary of findings, conclusions drawn from the study, and recommendations for policy and future research.

Citations — Why They Matter

A project without citations is an opinion, not research.

Every factual claim, statistic, definition, or idea that is not your own **must be cited**.

In-Text Citations (APA 7th Edition)

- **One author:**
(Mensah, 2021) or Mensah (2021)
- **Two authors:**
(Asante & Owusu, 2020)
- **Three or more:**
(Adjei et al., 2019)
- **Organisation:**
(Ghana Statistical Service [GSS], 2023)
subsequent: (GSS, 2023)
- **Direct quote:**
(World Bank, 2022, p. 45)

Reference List (End of Project)

- Every in-text citation **must appear** in the reference list
- Listed **alphabetically** by first author's surname
- Use a consistent style throughout: **APA**, Chicago, or Vancouver
- Software, datasets, and reports **must be cited**
- Use **DOI** where available

In-Text vs Reference List — Quick Guide

Type	In-Text (APA 7th)	Reference List
Journal Article	(Adjei & Boateng, 2021)	Adjei, K., & Boateng, A. (2021). Title of article. <i>Journal Name</i> , 10(2), 45–60. https://doi.org/xxx
Book	(Gujarati, 2022)	Gujarati, D. N. (2022). <i>Basic econometrics</i> (6th ed.). McGraw-Hill.
Govt. Report	(GSS, 2023)	Ghana Statistical Service. (2023). <i>Consumer price index report</i> . Author.
Software (R)	(R Core Team, 2023)	R Core Team. (2023). <i>R: A language and environment for statistical computing</i> (Version 4.3.1) [Computer software]. R Foundation.
Website	(World Bank, 2022)	World Bank. (2022). <i>Data: Inflation</i> . https://data.worldbank.org/

Using L^AT_EX for Citations — BibTeX/Biblatex

Why use BibTeX?

Automatic, consistent, error-free referencing. No manual formatting!

Step 1: Create a references.bib file:

```
@article{Adjei2021,  
  author = {Adjei, Kofi and Boateng, Ama},  
  title = {Inflation and household welfare in Ghana},  
  journal = {Journal of African Economics},  
  year = {2021},  
  volume = {30}, number = {2}, pages = {45--60}  
}
```

Step 2: In your .tex file add at the top:

```
\usepackage[style=apa, backend=biber]{biblatex}  
\addbibresource{references.bib}
```

Step 3: Cite inline: `\parencite{Adjei2021}` → (Adjei & Boateng, 2021)

Step 4: At the end add: `\printbibliography`

Kill the Fluff — Academic Language Matters

Banned Phrases in Academic Writing

The following constructions are **vague, lazy, or incorrect**. Eliminate them from your writing entirely.

Avoid This	⇒ Write This Instead
“According to a research...”	⇒ Name the source: “According to Mensah (2021) ...”
“A research by Wayo (2022) says...”	⇒ “ Wayo (2022) found that... ” or “...was established in a study on X (Wayo, 2022).”
“Many researchers have said...”	⇒ Cite at least two: “Several studies (Asante, 2020; Boateng, 2021) have shown...”
“It is believed that...”	⇒ <i>By whom?</i> “ Adjei (2019) argued that...”
“Studies show that...”	⇒ “ Empirical evidence (Owusu, 2022; IMF, 2023) suggests...”
“The study will benefit...”	⇒ Just say what the benefit is: “Findings will inform...”

The Golden Rule of Academic Citation Language

“A research” does not exist

- The word *research* is **uncountable** in English.
- You cannot say “**a research**” — it is grammatically wrong.
- Say: “**a study**”, “**research by...**”, or “**findings from...**”

Wrong: “A research done by Wayo (2022). . .”

Right: “A study by Wayo (2022). . .”

Right: “Wayo (2022) investigated. . .”

Right: “. . . as noted by Wayo (2022).”

Strong Reporting Verbs — Use These

Replace “says” or “talked about” with precise verbs:

- **For findings:** found, revealed, demonstrated, showed, established
- **For arguments:** argued, contended, maintained, posited
- **For descriptions:** described, outlined, identified, noted
- **For comparisons:** compared, contrasted, distinguished
- **For recommendations:** recommended, suggested, proposed

Important Reminders

Avoid These Mistakes

- **Plagiarism:** Always paraphrase and cite
- **Uncited claims:** No statement of fact without a reference
- **Inconsistent tense:** Use present tense for background; future or present for objectives
- **Objectives $\neq$ Methods:** Objectives say *what*, not *how*
- **Vague significance:** Always say *how* each party benefits
- **Missing group members:** All names and IDs on the cover page

Good Writing Habits

- Write clearly and concisely — avoid “big” words unnecessarily
- Use third-person language: “*The study investigates...*” not “*We looked at...*”
- Be consistent: if you abbreviate (e.g., GSS), define it on first use
- Number all pages, tables, and figures
- Proofread for grammar and spelling
- Ask your supervisor to review before submitting

Checklist Before Submitting Chapter 1

Structure & Content

- ☐ Cover page with all names & IDs
- ☐ Background flows broad → narrow
- ☐ Problem statement is specific & cited
- ☐ One clear main objective
- ☐ 3–5 specific, measurable objectives
- ☐ Significance addresses named beneficiaries
- ☐ Methodology has exactly 3 paragraphs
- ☐ Organisation covers all chapters

Referencing & Formatting

- ☐ All facts are cited in-text
- ☐ Reference list at end of project
- ☐ Consistent citation style (APA/Chicago)
- ☐ Software & datasets are cited
- ☐ Page numbers on all pages
- ☐ Tables/figures numbered and titled
- ☐ Acronyms defined on first use
- ☐ Proofread for grammar/spelling

Final Reminder

Chapter 1 sets the **tone** for your entire project. Invest time here — a strong introduction gives you a clear roadmap for everything that follows.

Summary — Chapter 1 at a Glance

Section	Purpose	Key Tip
Cover Page	Identify the project & group	Include ALL names & IDs
Background	Contextualise the study	Funnel from global to local; cite everything
Problem Statement	State the research gap	Specific, cited, 1–2 paragraphs
Objectives	Define what you will do	1 main + 3–5 specific; use action verbs
Significance	Justify the study	Name beneficiaries + say how they benefit
Methodology	Summarise your approach	3 paragraphs: data, methods, software
Organisation	Map the chapters	One sentence per chapter
References	Credit your sources	Every claim cited; consistent APA style

References I

-  Ghana Statistical Service. (2023). *Consumer price index: December 2022 report*. Author. <https://statsghana.gov.gh>
-  International Monetary Fund. (2023). *World economic outlook: A rocky recovery*. IMF. <https://www.imf.org>
-  R Core Team. (2023). *R: A language and environment for statistical computing* (Version 4.3.1) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org>
-  Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., . . . Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>
-  World Bank. (2023). *Inflation, consumer prices (% annual)—Ghana* [Data set]. World Bank Open Data. <https://data.worldbank.org>

Good Luck with Your Projects!

Remember:

*“A research project is a conversation with existing knowledge.
Cite your sources. Be precise. Be honest.”*

Questions?

See your supervisor or visit office hours.

Presentation prepared for Undergraduate Statistics Project Students
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